FINLIT VENTURES

Macro & Sector Outlook

Product Requirements Document (PRD) & Technical Implementation Specification

Purpose
A central macroeconomic intelligence engine that translates Indian macro data into explainable, sector-specific outlook scores displayed on every listed-company analysis page.

Document control	Details
Version	1.0
Product area	Individual Stock / Company Analysis
Feature	Macro & Sector Outlook
Primary market	Indian listed equities (NSE & BSE)
Audience	Software engineers, product, data and QA teams
Status	Implementation specification – initial model assumptions require validation

Product principle
Do not just show investors how a company is performing. Also show them how the economic environment surrounding that company is changing.

Document contents

- 1. Product objective and scope
- 2. Placement and user experience
- 3. Macro data engine and data governance
- 4. Indicator normalisation
- 5. Sector sensitivity model
- 6. Macro score calculation
- 7. Company-sector mapping
- 8. Historical intelligence
- 9. Database schema
- 10. API specification
- 11. Automation and refresh schedule
- 12. Admin panel
- 13. Error handling, security and performance
- 14. Acceptance criteria and QA test cases
- 15. Implementation roadmap
- 16. Future enhancements and launch checklist

Implementation note: This PRD defines the initial product and engineering requirements. Sector sensitivities, thresholds and model parameters are starting assumptions, not validated investment signals. Calibrate and backtest before exposing scores as reliable outputs.

1. Product objective and scope

Build a dedicated **Macro & Sector Outlook** section on every individual stock/company analysis page. The feature evaluates how the current Indian macroeconomic environment may support or pressure the sector in which a company operates.

The module must answer: (a) Is the macro environment favourable for this sector? (b) Which indicators are driving the assessment? (c) What external factors should investors monitor? (d) How has the outlook changed over time?

Out of scope: company-specific financial health, stock valuation, price targets, buy/sell recommendations and technical signals. The Macro Score must remain separate from Fundamental Score and Technical Score.

Core architecture: one shared macro dataset → normalised indicator signals → sector-specific sensitivity profiles → sector scores → company-page presentation.

2. Placement and user experience

Recommended placement: after Company Overview / Business Profile and before or alongside Fundamental, Technical and Valuation analysis. The section should be available on every supported company page.

Section heading: MACRO & SECTOR OUTLOOK.

Component A – Macro Backdrop Summary: compact horizontal summary showing sentiment, mapped sector, score out of 100, two most influential drivers and a “How it works” link.

Example: MACRO BACKDROP | Strong Headwind | Real Estate | 22/100 | Rate hikes (–), Credit (+). Values are illustrative only.

Component B – Sector Impact: comparative table showing sector, horizontal score bar, score, sentiment and top reasons. Highlight the currently viewed company’s sector. Sort descending by score by default; allow expansion to all supported sectors. Include a visible midpoint marker at 50.

Component C – Indicator Signals: grid grouped by Inflation, Monetary Policy, Credit & Banking, Economic Growth, Currency and Commodities. Each tile shows latest value, previous value/change, direction, normalised signal, observation date and source.

Component D – Sector Interpretation: concise explanation generated from the largest actual score contributions. Show both supporting and adverse factors, plus relevant indicators to monitor.

Component E – How It Works: expandable methodology covering score meaning, indicators, normalisation, sensitivities, formula, sources, update frequency and limitations.

Score	Sentiment
80–100	Strong Tailwind
65–79	Tailwind
45–64	Neutral
25–44	Headwind
0–24	Strong Headwind

Visual design: restrained colours, strong readability, compact rows, subtle badges, neutral midpoint, responsive desktop/mobile layout. Suggested palette: dark green for strong positive, muted green for positive, grey for neutral, muted red for negative and dark red for strong negative. Avoid excessive cards and colour saturation.

3. Macro data engine and data governance

Create a central ingestion and scoring service. Do not fetch macroeconomic data independently for each company page. Store raw readings centrally and calculate sector scores once per data/model snapshot; company pages consume cached results.

Every raw observation must retain indicator code, value, unit, observation date, publication date, source name/URL, frequency, data status, retrieval timestamp and revision/version metadata.

Indicator	Preferred source	Expected cadence
CPI and Food CPI	MoSPI	Monthly, after official release
WPI	Office of Economic Adviser	Monthly, after official release
GDP growth	MoSPI	Quarterly, after official release
IIP	MoSPI	Monthly, after official release
Repo rate / policy actions	RBI	On announcement
Bank credit / deposits	RBI	On official release
USD/INR	RBI or validated market-data provider	Daily
Brent crude	Validated commodity-data provider	Daily
Unemployment	Official or validated statistical source	As published
Other commodities	Validated source relevant to sector	Daily/monthly as available

Data validation: check presence, valid observation/publication dates, units, acceptable ranges, duplicates, freshness and source revisions. Status values should include Available, Stale, Missing, Invalid and Revised.

Publication-date rule: distinguish observation date from publication date. The engine must not use information before it was publicly available at the score's as-of timestamp. This avoids look-ahead bias in historical scores.

Missing-data rule: never silently replace missing values with zero or neutral. Exclude unavailable inputs, calculate coverage and apply the minimum-coverage rules in Section 6. Serve the last valid snapshot with a stale-data warning if a refresh fails.

4. Indicator normalisation engine

Raw macro indicators use different units and directions. Convert each active indicator into a normalised signal between –1 and +1 before applying sector sensitivities. Higher values are not universally better; direction depends on the indicator and sector context.

Two-part signal: calculate a Level Signal and a Momentum Signal, each capped to [–1, +1].

```
Indicator Signal = (Level Signal + Momentum Signal) / 2
```

Level signal evaluates whether the current reading is favourable relative to configured thresholds, target ranges or historical benchmarks. **Momentum signal** evaluates whether the indicator is improving or deteriorating over a configured comparison period.

Supported normalisation methods: threshold/range-based, historical percentile, percentage-change, rate-of-change/momentum and capped z-score methods. Each indicator must specify its method, lookback window, direction, thresholds, units and treatment of extreme values in configuration.

Examples: inflation may be favourable when easing from an elevated level but adverse if rising sharply; credit growth may be positive up to a healthy range but excessive growth can indicate overheating; repo-rate changes must be distinguished from the current repo-rate level.

5. Sector sensitivity engine

Create a configurable sector-indicator sensitivity matrix. Each sector has a sensitivity coefficient for each relevant indicator, between –1.00 and +1.00. Coefficients represent expected direction and strength of sector sensitivity; they are model assumptions and do not prove causality.

Coefficient	Interpretation
+1.00	Strong positive relationship
+0.75	High positive relationship
+0.50	Moderate positive relationship
+0.25	Low positive relationship
0.00	No material relationship
–0.25 to –1.00	Increasing degrees of negative relationship

Illustrative sector profiles (not production-calibrated):

Sector	Potential positive drivers	Potential negative drivers
Private banks	Credit demand; healthy funding; selected rate conditions	Funding cost pressure; rising NPAs
PSU banks	Credit growth; asset-quality improvement	Funding pressure; rising NPAs
NBFCs / HFCs	Credit demand; funding availability	Higher borrowing costs; liquidity stress
Real estate	Lower rates; credit availability; income growth	Rate hikes; construction-cost inflation
Automobiles	Lower rates; consumer demand; credit	High borrowing costs; weak consumption
FMCG	Consumption; healthy real income	Food and input-cost inflation
Cement	Infrastructure and construction demand	Energy, freight and financing costs
Capital goods	IIP; public capex; credit growth	High financing costs
Metals & mining	Industrial demand; output prices	Falling prices; energy costs
Specialty chemicals	Industrial demand; favourable input-output spread	Input-cost inflation
Paints	Demand; lower crude-derived input costs	Crude and raw-material inflation
Aviation	Passenger demand; lower fuel prices	Crude prices; currency depreciation
Oil marketing	Marketing margins; favourable crude conditions	Inventory losses; adverse pricing
Upstream oil & gas	Higher realised crude prices	Falling realised crude prices
Pharma	Export demand; favourable currency	Input costs; regulatory issues
Hotels & QSR	Travel; discretionary demand	Food inflation; weaker consumption
IT services	Global technology demand; favourable currency	Global demand slowdown
Agri inputs	Agricultural activity; farm economics	Adverse input/commodity conditions

Sensitivity profiles must be stored in the database/configuration service, not hardcoded in frontend logic. Include effective dates, active status, model version and audit trail. Validate that sector-specific drivers are economically plausible. For banks, distinguish asset yields, deposit repricing, funding costs and credit quality rather than assuming rate hikes are always positive or negative.

For diversified companies, use segment-level exposures where reliable revenue contribution data exists. Avoid applying one broad sector coefficient to materially different business models.

6. Final Macro Score calculation

Step 1 – Indicator signal: compute the average of level and momentum signals, each capped between –1 and +1.

```
IndicatorSignal■ = (LevelSignal■ + MomentumSignal■) / 2
```

Step 2 – Sector contribution: multiply indicator signal by sector sensitivity. If a separate weight is configured, include it explicitly.

```
Contribution■ = IndicatorSignal■ × Sensitivity■ × Weight■
```

Step 3 – Aggregate: sum valid contributions. Use a consistent weight normalisation policy. Recommended: divide the sum by the sum of absolute active weights for the valid indicators, then scale to the model's configured signal range. The chosen method must be fixed, versioned and documented.

Step 4 – Convert to score: apply a bounded tanh transformation.

```
MacroScore = 50 + 50 × tanh(WeightedMacroSignal / K)
```

K is a configurable normalisation factor. An initial trial value of 3 may be used for calibration, not assumed correct. Score is bounded between 0 and 100; a zero aggregate signal produces 50.

Important implementation clarification: the earlier illustrative formula used a simple sum divided by 3. For production, the engineering team must implement one explicitly defined aggregation method (including weight normalisation) and validate the resulting distribution on historical data. Do not mix raw sums and normalised weighted averages.

Coverage: calculate valid active weight divided by total active weight. Recommended initial minimum coverage is 70%, subject to validation. Also define critical indicators per sector; a sector must not receive a confident score if a critical input is missing even when aggregate coverage passes. If coverage fails, return "Insufficient Data" rather than a fabricated score.

Top drivers: rank valid indicators by absolute contribution. Display the two largest, with sign, latest value and unit. If contributions are close, use a deterministic tie-break rule such as configured priority, then freshness.

Sentiment bands: 80–100 Strong Tailwind; 65–79 Tailwind; 45–64 Neutral; 25–44 Headwind; 0–24 Strong Headwind.

7. Company-sector mapping

Maintain a standard sector master and company-to-sector mapping table. Initial coverage should include: private banks, PSU banks, NBFCs, housing finance, insurance, real estate, automobiles, auto ancillaries, FMCG, consumer durables, cement, capital goods, infrastructure, metals & mining, specialty chemicals, paints, pharmaceuticals, hospitals, IT services, telecom, aviation, hotels & QSR, oil marketing, upstream oil & gas, power generation, power transmission, renewable energy, agri inputs, textiles, logistics, media & entertainment, paper, sugar, fertilisers and diversified.

Mapping must be editable by authorised administrators and retain effective dates. Every listed company should have a primary sector mapping. Where segment revenue data is available, use a revenue-weighted company score:

```
CompanyMacroScore = Σ (SegmentRevenueWeight■ × SectorMacroScore■)
```

Segment weights must total 100% (or be normalised with an explicit rule). If segment data is unavailable, use the primary business sector and disclose that limitation. Do not invent segment weights.

8. Historical macro intelligence

Persist immutable score snapshots so users can view score history and reproduce past calculations. UI ranges: 1M, 3M, 6M, 1Y and All. Show current score, changes over selected periods, line chart, 50 midpoint reference and dates.

Store score snapshot, input observations, contributions, sensitivity configuration version, model version and as-of timestamp. If source data is revised, retain the original published snapshot. Optionally calculate a separately labelled revised historical series; never silently overwrite history.

9. Database schema

Table	Purpose / key fields
macro_indicators	Current raw readings: indicator_code, value, unit, observation_date, publication_date, source, frequency, status.
macro_indicator_history	Time series and revisions: indicator_code, value, observation_date, publication_date, revision_number.
macro_indicator_signals	Normalised signals: level_signal, momentum_signal, final_signal, calculation_date, coverage.
sector_master	Sector catalogue: sector_code, name, description, active status.
sector_sensitivity	Sector-indicator settings: sector_id, indicator_code, coefficient, weight, normalisation_method, effective dates.
company_sector_mapping	Company and segment mapping: company_id, sector_id, revenue_weight, primary flag, effective dates.
sector_macro_scores	Score snapshots: sector_id, score, label, weighted_signal, coverage, top drivers, model version, snapshot_date.
sector_macro_contributions	Per-score inputs: indicator value, signal, sensitivity, weight, contribution and direction.
macro_model_configuration	Model settings: K, coverage thresholds, score bands, model version, effective date, active flag.

Use decimal/numeric types for values, scores and coefficients. Add foreign keys, indexes on indicator_code, sector_id, company_id and snapshot_date, and uniqueness constraints to prevent duplicate observations/snapshots. Preserve source provenance and configuration versions.

10. Backend API specification

Method and endpoint	Purpose / response
GET /api/v1/companies/{company_id}/macro-outlook	Company mapping, sector score, sentiment, coverage, top drivers, indicator summary, as-of date and model version.
GET /api/v1/macro/sector-impact	All available sector scores, labels, drivers and updated timestamp; support sort/filter.
GET /api/v1/macro/sectors/{sector_id}/indicators	Indicator values, prior values, signals, sensitivities, contributions, sources and dates.
GET /api/v1/macro/sectors/{sector_id}/history?period=3M	Historical score snapshots for 1M/3M/6M/1Y/ALL.
GET /api/v1/macro/methodology	Definitions, formula, score bands, sources, model version and limitations.
POST /api/v1/admin/macro/sector-sensitivity	Authorised admin update of coefficients; validate and audit every change.

API responses should include explicit units, observation/publication dates, data status, coverage, calculation timestamp and model version. Return structured errors for unavailable company mappings, stale data and insufficient coverage. Public endpoints should be read-only.

11. Automation and refresh schedule

Recommended cadence: USD/INR and Brent daily; repo rate on RBI announcements; CPI, WPI and IIP monthly after official release; GDP quarterly; credit/deposit growth on official release; unemployment as published. Refresh schedules must follow source publication calendars rather than assumed dates.

Pipeline: fetch → validate → persist raw observation → detect revisions → normalise signals → apply sector sensitivities → calculate scores → identify drivers → generate explanation → save immutable snapshot → invalidate/refresh cache.

Trigger recalculation when a relevant new observation is published or when an authorised administrator activates a new model/sensitivity version. Avoid recalculating all companies independently.

12. Admin panel requirements

• Add, edit, deactivate indicators and configure units, frequency, source and normalisation method.

• Update sector-indicator sensitivity coefficients, weights and critical-indicator flags.

• Configure score bands, normalisation factor K and minimum coverage rules.

• View missing, stale, invalid and revised source data.

• Manually trigger ingestion, validation and score recalculation.

• Edit company-sector and segment mapping.

• Inspect indicator contributions and score calculation logs.

• Review model versions, effective dates and historical snapshots.

• Audit every configuration change with user, timestamp, old value and new value.

Admin configuration must be role-protected, validated and versioned. A model change should create a new effective version; it must not silently alter previously published snapshots.

13. Error handling, security and performance

Error cases: missing data, stale data, source outage, invalid units, duplicate observations, revised data, newly listed companies, diversified companies without segment data, insufficient sector coverage, extreme readings, conflicting indicators and failed score jobs.

Fallback: serve the last successful snapshot with its original timestamp and a visible stale-data warning. Do not display a stale score as current.

Security: only authorised admins can change model settings. Public endpoints are read-only. Validate inputs, apply rate limits where appropriate, store secrets securely and audit administrative changes.

Performance: cache sector scores and indicator payloads; do not call external macro sources during company page rendering. Keep the macro engine independent from fundamental and technical scoring services.

14. Acceptance criteria and QA test cases

ID	Acceptance requirement
AC01	Correct company-sector mapping is shown.
AC02	Raw macro observations retain source, units and both relevant dates.
AC03	Normalised indicator signals remain between –1 and +1.
AC04	Correct sector-specific sensitivity and weight are applied.
AC05	Score remains between 0 and 100 and formula is reproducible.
AC06	Correct sentiment band is assigned.
AC07	Top drivers are selected by absolute contribution.
AC08	Sector comparison table shows all supported sectors and highlights current sector.
AC09	Indicator breakdown displays actual values, units, signals, dates and sources.
AC10	Historical snapshots are retrievable and immutable.
AC11	Missing/stale data is clearly flagged; no artificial neutral values.
AC12	Diversified companies support revenue-weighted sector exposure.
AC13	Admin can update sensitivities without a code deployment.
AC14	Model version and calculation inputs are auditable.
AC15	Responsive UI works on desktop and mobile.
AC16	Company page does not independently fetch macro data from external sources.

QA scenarios

Scenario	Input	Expected behaviour
Real estate	Rising rates; strong credit; higher construction costs	Rate and construction-cost contributions negative; credit contribution positive; aggregate calculated from valid weighted inputs.
Private banks	Healthy credit; changing repo; deposit growth changes	Credit positive if configured; funding and repricing effects evaluated separately.
FMCG	Food/input inflation rises; consumption weakens	Cost pressure negative; demand signal separately assessed.
Aviation	Crude rises; rupee depreciates; passenger demand improves	Fuel and FX pressures negative; demand may be positive.
Missing data	CPI missing; credit and GDP available	CPI excluded; coverage recalculated; score only if coverage and critical-input rules pass.
Zero contributions	All valid contributions equal zero	Score equals 50; label Neutral, provided coverage is sufficient.
Source outage	Latest refresh fails	Serve last successful snapshot with explicit stale warning.
Revised data	Source revises a prior reading	Retain original published snapshot; optionally create labelled revised series.

15. Implementation roadmap

Phase	Deliverables	Exit condition
1. Foundation	Database, source integrations, validation, raw history	Reliable observations with provenance and dates.
2. Scoring engine	Normalisation, sensitivity matrix, score calculation, mapping, drivers	Reproducible sector scores with unit tests.
3. Frontend MVP	Backdrop bar, sector table, indicator grid, explanation, methodology	Company pages render accurate cached results.
4. Historical intelligence	Snapshots, score history chart, model versioning	Historical scores reproducible.
5. Admin and QA	Admin controls, audit logs, monitoring, regression tests	Safe configuration and production readiness.

MVP priority: data ingestion, normalisation, sector sensitivity, scoring, company mapping, backdrop bar, sector table, indicator grid and methodology. Historical charts and admin tooling should follow, but configuration/versioning and auditability must exist before production scores are published.

16. Future enhancements and launch checklist

Future enhancements (not MVP blockers): macro score versus sector-index performance; macro event impact; sector rotation intelligence; scenario analysis (e.g. repo ±25 bps, crude ±10%, CPI +1 percentage point); and threshold-based macro alerts.

Launch checklist:

• All data sources approved and documented; licensing/usage rights checked.

• Observation date and publication date stored separately.

• Normalisation direction and thresholds reviewed by product/data owner.

• Sector sensitivities reviewed and backtested; model limitations disclosed.

• Aggregation and missing-data policy finalised and unit-tested.

• Historical snapshots immutable and reproducible.

• Company mapping and segment weights reviewed.

• UI displays as-of date, coverage, source and stale-data warnings.

• Admin access, audit trail, monitoring and rollback tested.

• Investment-advice disclaimer and methodology explanation reviewed.

Final product outcome: Every supported stock page should connect Macroeconomic Data → Indicator Signals → Sector Sensitivities → Sector Macro Score → Investor Interpretation. The feature should be scalable, transparent, explainable and independently maintained from Fundamental and Technical scoring.

Disclaimer: The Macro & Sector Outlook is an informational model of sector-level macroeconomic conditions. It is not a company-specific assessment, forecast, or investment recommendation. Sensitivity assumptions and model outputs are subject to uncertainty and must be validated.